University of Southern California

Viterbi School of Engineering
Aerospace and Mechanical Department

AME 546: Design for Manufacturing & Assembly

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Final Project : Hand Grinder

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Introduction:

A hand or angle grinder, a commonly used mechanical tool, comprises several intricately designed parts that work together to achieve manual grinding as shown in figure 1. These parts typically include the handle, shaft, burrs (or grinding elements), housing, fasteners, and locking mechanisms. In its conventional form, a hand grinder can involve components and design features that are not optimized for cost-effective manufacturing or efficient assembly. Through this project, each component of the grinder will be critically analyzed and re-evaluated under the DFM and DFA framework to identify areas for improvement.

This project focuses on the comprehensive evaluation and redesign of a hand grinder using the principles of Design for Manufacturing (DFM) and Design for Assembly (DFA). The objective is to optimize each individual component and the overall assembly process of the hand grinder to enhance manufacturability, reduce complexity, and minimize production costs while maintaining the intended functionality and durability of the product.

Figure 1- Hand/angle grinder.

The DFM approach will be used to examine how each part can be manufactured with minimal cost, waste, and processing time. Material selection, tolerances, geometry, and machining processes will be reviewed to determine potential simplifications and design alterations. For example, part features that are difficult or expensive to machine will be redesigned for ease of production, and material choices will be re-evaluated for both performance and cost-efficiency.

Concurrently, DFA principles will be applied to streamline the assembly process. This will involve reducing part count, simplifying joining methods, and ensuring ease of alignment and handling during assembly. The goal is to reduce the total assembly time and associated labor costs, which are often significant contributors to the final product price.

A cost estimation analysis will be performed for each component individually as well as for the complete assembly. This includes direct manufacturing costs, material costs, and labor-related expenses for assembly. Comparisons will be drawn between the original and optimized designs to quantify the improvements achieved through the DFM and DFA application.

This project not only aims to demonstrate the practical benefits of DFM and DFA in a real-world product but also develops a methodology that can be extended to a wide range of consumer mechanical products to enhance competitiveness and sustainability in manufacturing.

Product details:

The exploded view of the angle grinder reveals a detailed layout of all internal components and their assembly relationships. This view is essential for understanding the tool's mechanical structure, aiding in design analysis, maintenance planning, and manufacturing strategy. The assembly can be broadly categorized into four functional sections: the motor, gear transmission, tool interface, and outer housing with control components.

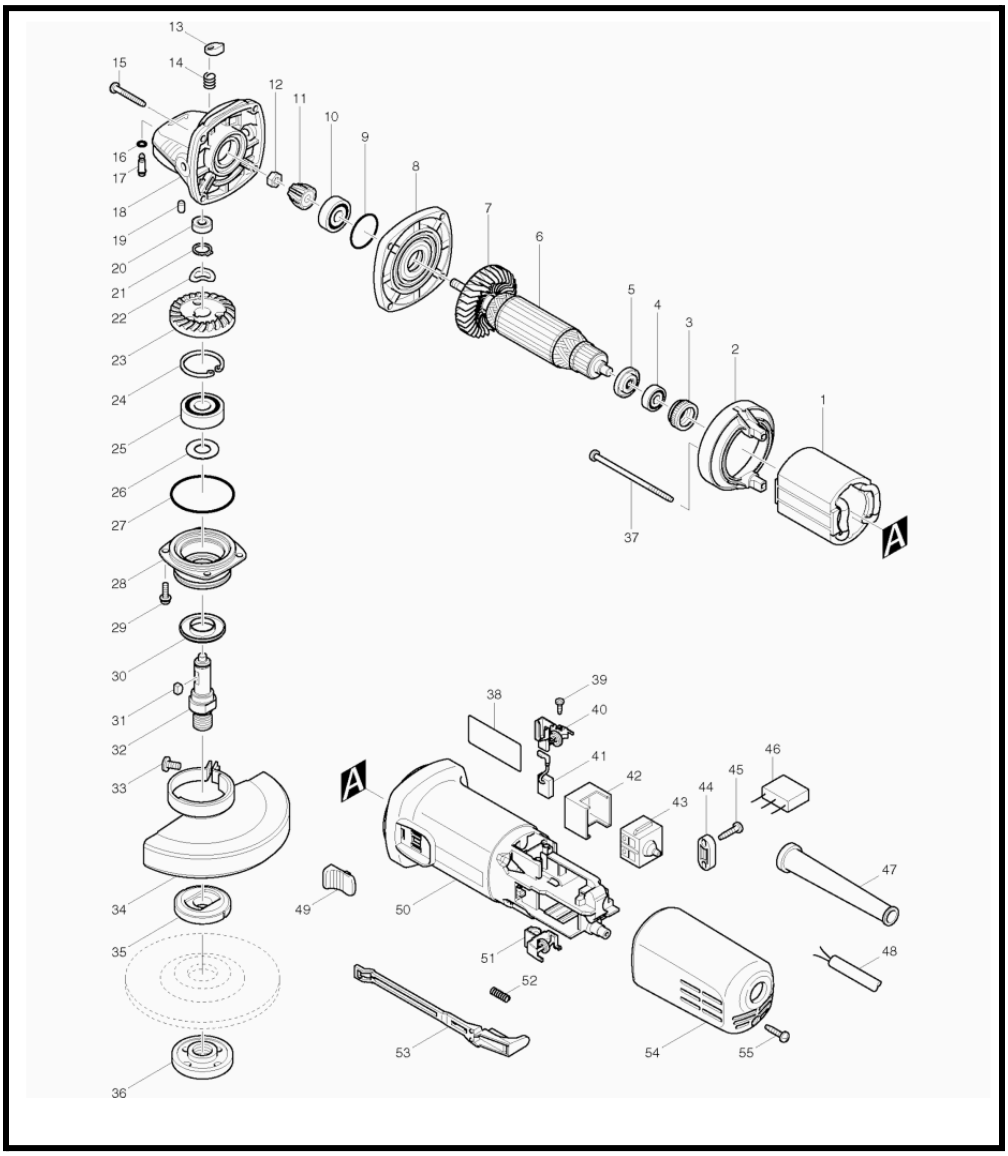

Figure 2- Exploded view of Makita hand grinder

Starting from the right side of the figure 2, the motor section consists of the field coil (stator) and the armature assembly. The field coil is mounted inside the motor housing and generates a magnetic field, while the armature (which includes a shaft and commutator) rotates within this field to produce mechanical power. The armature is supported at both ends by high-precision ball bearings and isolated using insulation washers. This arrangement ensures smooth and electrically safe operation at high rotational speeds. A long pin is shown securing the armature axially within the housing.

Moving toward the center, the gear transmission section includes the gear housing cover, a bearing arrangement, and the core gear housing made from aluminum alloy. Inside this housing, the power is transferred from the armature to the spindle via a pair of spiral bevel gears. Proper meshing and alignment of these gears are essential for torque transmission without vibration or noise. Supporting this assembly is the bearing box, which holds the output bearing and provides rigidity to the entire front-end structure.

The tool interface section includes the spindle shaft, which is directly driven by the gear set. The grinding disc is mounted on this spindle and is clamped between an inner flange and a lock nut. Over this setup is the wheel cover (guard), a curved steel shield that protects the user from sparks, debris, and accidental contact with the rotating disc. This cover is typically adjustable and may be removed or repositioned depending on the task.

The outer housing and control section, located on the left side of the diagram, consists of the motor housing, electrical wiring, carbon brushes, and switch components. The plastic housing provides insulation and ergonomic handling. The brushes transfer power to the commutator, while the switch activated by a plastic lever—controls the tool's on/off function. A side handle is also provided to give the user additional grip and control. The power cord exits the rear of the tool and is supported with a strain relief to prevent internal wire damage.

This exploded view reflects a modular and serviceable design philosophy. Fasteners are primarily self-tapping screws, enabling quick assembly. The use of standard components like bearings, washers, and fasteners helps streamline production, while custom components like the spindle, gear housing, and motor parts are likely manufactured in-house or closely controlled through dedicated suppliers. Overall, the design emphasizes durability, precision alignment of moving parts, and operator safety.

Cost Analysis for Non - Standard Part:

1. **Spiral bevel gear and pinion:**

Spiral bevel gears in an angle grinder are used to change the direction of rotation and transfer power between shafts that are typically at a 90-degree angle. They are located within the angle grinder's gearbox, specifically engaging with the motor's output shaft and the grinding wheel's spindle. The spiral shape of the gear teeth ensures smooth and quiet operation, reducing vibration and noise compared to straight bevel gears. When the motor's shaft turns, the spiral bevel gear meshing with it rotates, and this rotation is then transmitted through another spiral bevel gear connected to the grinding wheel's spindle. Spiral bevel gears offer advantages like smoother operation, reduced noise and vibration, and higher load-carrying capacity compared to straight bevel gears, making them suitable for high-speed applications like angle grinders.

Figure 3- Spiral bevel gear and pinion.

The dimensions of the gears are shown in figure 3. The number of teeth on spiral bevel gear is 36 while the pinion has 11 teeth, making the gear ratio of 3.27. This means that the available speed at the spindle of the angle grinder is 3.27 times more than the motor output speed. The outer diameter (OD) of the gear is 47.5 mm with a bore of 12 mm and keyway of 4mm. The pinion has a thickness of 15 mm and bore of 6mm as shown in the figure 3. The material used is high Carbon Alloy Steel-AISI 4140. The manufacturing of gears will include several processes like machining (to bring raw materials size to part dimension), gear hobbing, heat treatment and finishing (grinding) operations. The machining cost includes material cost, machining time (volume of material to be removed, difficulty of machining), tool cost, setup time and labor cost.

<u>Cost estimation:</u>

Assuming material cost to be $5/ part.

Calculating rough turn, finish turn, drilling, gear cutting, broaching, grinding costs:

Setup time= 0.65 s (from textbook table 6.4)

Load and unload time= 23.3 s

Total positioning time= 1.5s

lw= 30mm or 1.18 inch, da= 50 mm or 1.96 inches, db= 48 mm or 1.88 inches

Vm= 0.284 in^3 ρs=1.3 for alloy steel, Pm= 8 hp, Vf= 112.32 from textbook tables 7.6

tmp= 2.77 s, tmc= 3.72 s, Am=6.96 in^2

™'= **5.81 s**

Similarly for finish turn

™'= **6s**

For drilling:

Setup time= 1s, positioning time= 9s, load and unload time= 23.3s

The total machine time for rough and finish drilling came out to be ™'= 5.35+3=**8.35s**

Gear cutting:

Setup time= 0.9 hr, load and unload time= 41.3s (4 jaw), positioning time= 39s

Length of the gear cut= 24.14 in

Assuming it takes 75 s to cut one inch then

Total machining time= 75*24.14=**1810.5s**

Broaching:

Setup time= 0.6s, positioning time= 13s, load and unload time= 41.3s

Total length= 1.37 inch

Assuming it takes 20 s to cut

Total machining time= 20*1.37=**27.4s**

Grinding:

Setup time= 0.6 hr, positioning time= 19 s, load and unload time= 41.3s

Total material to be grinded came out to be= 2.16*10^-3

Assuming 100 s then total machine time= 0.216

Total machining time= 11.81+5.35+3+1810.5+27.45+0.216= 1858.33 s

Assuming standard machine rate of $30/hr to avoid complexity in the calculations,

Total cost= 1858.33/3600*30 = $15.49

This cost includes both the gear and the pinion. Given that we have to manufacture 100,000 parts and hence assuming material cost per part to be $3. This cost is purely assumed and actual cost may change depending on the current material rate. The spiral gears will be given heat treatment to relieve stresses which are generated during the manufacturing processes and hence an additional cost of $0.75 is assumed per part. Therefore the total cost of the spiral gear was estimated to be **$19.24**.

2. Gear housing:

The gear housing of the angle grinder is a key structural component that encloses the spiral bevel gears, supports the spindle and bearings and transfers the motor's power to the grinding disc or spindle. The gear housing ensures reliable power transfer, minimizes wear and overheating, and plays a major role in the durability, safety, and efficiency of the tool. The key dimensions of the part are shown in figure 4 and the depth or height of the part is 50 mm.

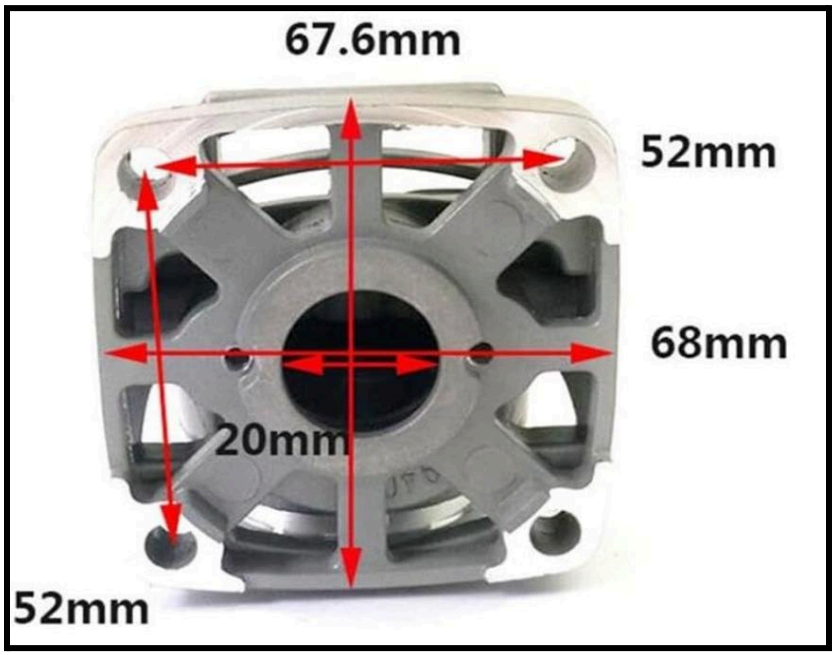

Figure 4 - Gear housing.

The material of the gear housing is die cast **Aluminum alloy-A13**. It is lightweight and easy to cast, has good thermal conductivity for heat dissipation, corrosion-resistant, and sufficient strength for supporting gears and bearings. The manufacturing process used for making the gear housing is **Die Casting**. Other manufacturing processes will include trimming and finishing processes. Alternative to die casting, we could use sand casting but considering we have to manufacture 100,000 parts per year and therefore it is not economical as die casting is more beneficial for mass production.

<u>Cost estimation:</u>

L= 68mm W=68mm and height= 50mm

Material- A13, A_{pc}= 55 cm^2

A_{pt}= 1.75*55=96.25 cm^2

f= 264 KN, Assuming the mass of the part to be 600 g.

V_p=600/2.66= 225.56 cm^3

V_s=225.56(1+⅔)= 375.93 cm^3, Ls= (2*5)+12=22 cm

From textbook table 10.4, selecting 1800 kN machine which satisfies all our requirements.

Ladle time= 1.80 s, Injection time= 0.028s, Cooling time= 5.71s, extraction time= 11.88s, lubrication time= 12.7s, reset time=4.9s

Cycle time= 1.8+0.028+5.71+11.88+12.7+4.9= 37.018s

From the machine data, machine rate is $73/h and 85% efficiency

Process cost= **$0.88**

Trimming time and cost:

tp= 29.88 s

Cost= **$0.39**

Material cost= **$1.12**

Total cost= 1.12+0.39+0.88= **$2.39**

Now, calculating the cost of die

Md= 62.5+0.029 $A_c h_p^{0.4}$

Clearance= 7.7 cm, Width of the plate= 8.16 cm, added width= 12.5cm

Plate area= 492.84 cm^2

h_p= 20.4 cm

Md= 110.24 hours

X_i= 0.59

M_x= 3.73 hours, M_e= 23.17 hours, M_{po}= 19.27 hours

Total manufacturing hours= 19.27+3.73+23.17+110.24=156.41 hours

Assuming standard machine rate of $40/hr

Die making cost= 156.41*40=$6256.4

Now, calculating the cost of trim die:

X_p=2, M_{to}= 15.04 hours, flw=1.58, M_t= 49.76 hours

Trim die cost= 49.76*40=$1990.4

Cost per part= (6256.4+1990.4)/100000=$0.082

Total cost per part= 2.39 + 0.082=**$2.472**

3. **Motor housing:**

The motor housing of the angle grinder is a critical component that encases and protects the motor assembly. It protects the motor from dust, debris, and mechanical damage, supports internal components, including stator, rotor, and cooling fan, maintains alignment between the motor shaft and gear mechanism and also assists in cooling by integrating ventilation slots and guiding airflow.

It ensures safe and reliable operation of the tool, extends motor life by shielding it from external contaminants, helps in heat dissipation, and prevents overheating, provides the structural base for mounting external switches, handles, and covers. The dimensions of the motor housing are shown in the figure 5 as below:

Figure 5 - Motor housing.

The material used is **polyamide (66 nylon)**. The cost per kg of the material is $4. Polyamide is heat resistant, light weight, corrosion resistant, good electrical insulation, and has good impact resistance. The manufacturing process used to manufacture this part is **Injection Molding** due to its highly complex shape and geometry and is suitable for mass production.

Cost calculation:

The total cost per part of the motor housing came out to be **$14.1** as per CustomPart.net. We assumed the wall thickness to be 2.5 mm. The machine has a clamping force of 440 tons and an hourly rate of $60 with single cavity mold. The cost per part and other factors such as projected area, projected volume is shown in the figure 6.

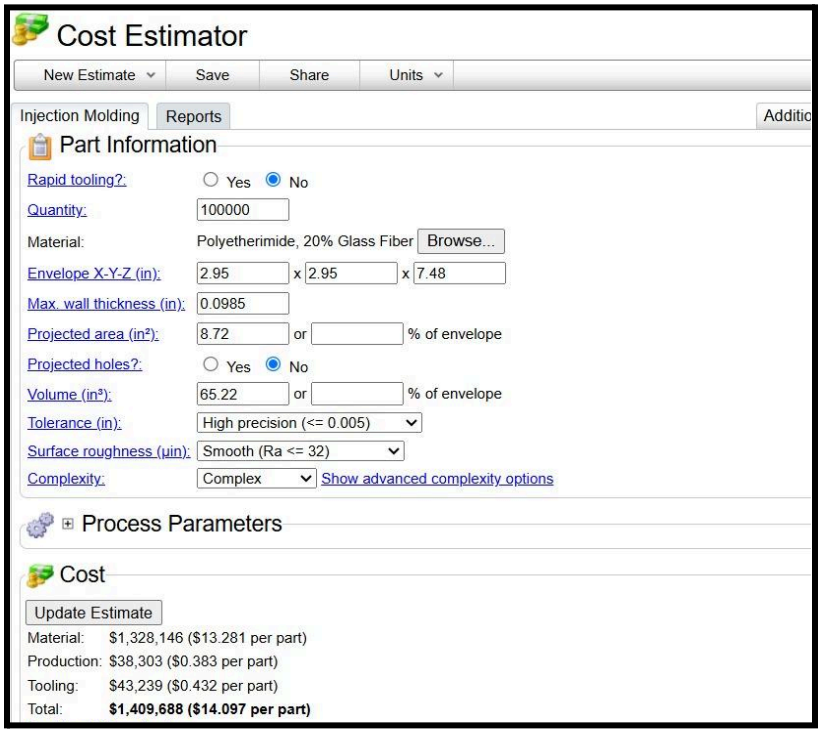

Figure 6 - Motor housing cost.

4. **Fan:**

The fan in the angle grinder is a small but essential component that plays a key role in cooling the motor during operation. The fan is mounted on the motor shaft and serves to:

- **Cool the motor** by drawing air through the tool during operation.
- **Expel dust and debris**, aiding in the protection of internal components.
- **Enhance airflow** through the grinder's labyrinth construction, which is designed to seal and protect the motor and bearings from contamination.

Figure 7- Fan

The material of the fan is **Polypropylene**. The cost of the material is $1.00 per pound as per the Custompart.net. The manufacturing process used is **Injection Molding** due to accuracy and precision of the manufacturing process. The diameter of the fan is 56 mm, wall thickness is 2 mm and height of the part is 5mm.The manufacturing cost of the fan came out to be $0.455 as per Custompart.net.

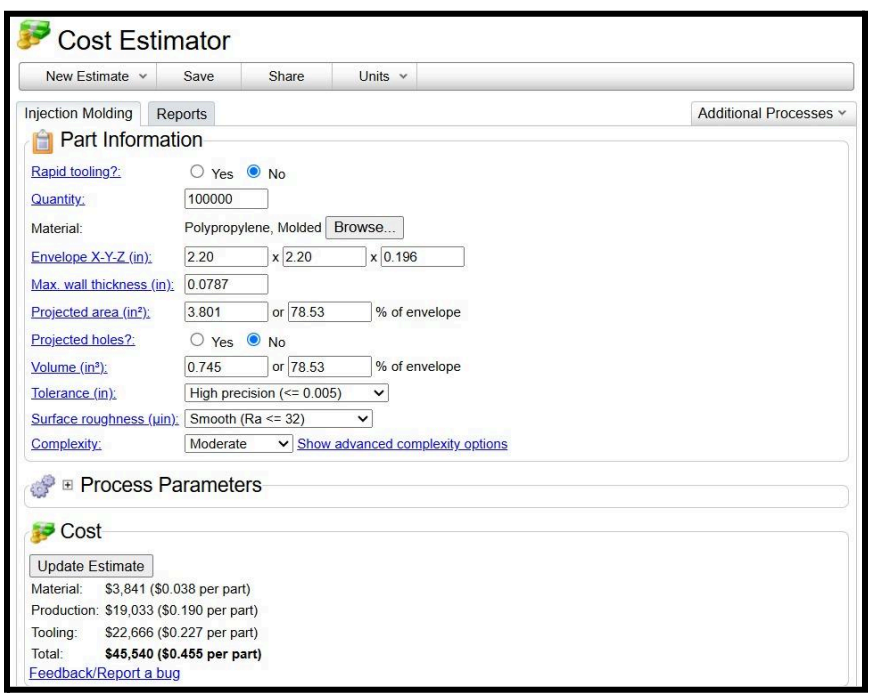

Figure 8 - Cost of the fan.

5. Bearing Box

The bearing box used in the angle grinder is typically made from **aluminum alloy (Aluminum A360, die cast)**. This material choice balances strength, weight, and thermal conductivity, making it ideal for housing components like bearings that require precise alignment and heat dissipation. This component is designed to house and support the internal bearings and spindle assembly. While the external dimensions provide a general idea of its size, the critical internal features such as bearing seat diameters, bolt hole patterns, and alignment surfaces are essential for proper function and typically require precise manufacturing tolerances. The process used for manufacturing the bearing box is **Die Casting** due to the material being suitable and the process is cheaper for a large number of units (100,000 units) in this case.

Figure 9: Bearing box part

<u>Cost Estimation</u>:

The cost estimation for the bearing box used in the angle grinder was carried out in the custom parts die casting estimator for 100,000 units annually, this gave a rough estimate on how much the manufacturing cost was required in making the bearing box. The selected material for this component is Aluminum A360.0, a commonly used alloy in die casting due to its excellent fluidity, dimensional stability, and good corrosion resistance. The part envelope was set to 6 × 4 × 2 inches (15.2 × 10.2 × 5.1 cm), based on known part dimensions and visual assessment from manufacturer references. The maximum wall thickness was set to 0.2 inches, which is a standard for structural parts requiring strength without excessive mass or cooling time. The volume of the part, calculated from its known weight of 40 oz (1134 g), was approximately 25.3 in^3, which aligns with the density of aluminum and confirms the accuracy of the dimensional assumptions.

The bearing box includes one central through-hole for the bearing mount and for projected holes with a quantity of 1 (central bore). A projected area of 24 in^2 was entered, representing the top face area that dominates mold filling. Given the nature of the component, which houses a bearing and interfaces with other high-precision parts in the gear system, a high tolerance value of less than 0.005 inches was

selected. The surface roughness was set to "Mechanical quality," appropriate for a functional component that does not require cosmetic finishing. The overall complexity was rated as simple due to its internal ribs, recessed features, and stepped profiles, although the part still remains moldable with a simple 2-part die with no side actions or side cores.

After giving all the required inputs, the resulting cost breakdown gave part price to be roughly $6.755 per part in material cost, $0.650 for production (casting process), and $0.620 in tooling amortization, totaling $8.025 per part close to final cost of $10.39 per part (ebay). This brings the total estimated cost for 100,000 units to approximately $802,478. These values reflect a realistic estimate for mass production of a precision-cast aluminum housing component, balancing accuracy, manufacturability, and functional design.

Material:	$675,465 ($6.755 per part)
Production:	$65,010 ($0.650 per part)
Tooling:	$62,002 ($0.620 per part)
Total:	**$802,478 ($8.025 per part)**

Figure 10: Cost estimate for bearing box

6. **Brush Holder:**

The brush holder is a small, non-rotating component typically made of a thermoplastic or thermoset polymer, designed to house the carbon brush that maintains electrical contact with the commutator of the motor. It features internal cavities or channels to retain the brush, provide spring tension, and accommodate electrical leads or connectors. It must also resist heat, electrical arcing, and mechanical vibration during operation.

The brush holder in the angle grinder is most commonly made of **PA66 (Nylon 66, 30% glass fibre filled)**, for added strength and thermal stability. This material is widely used in power tools because it offers excellent electrical insulation, heat resistance, and mechanical durability, making it ideal for holding carbon brushes under spring pressure in a high-vibration environment.. Given its intricate shape and the need for precise dimensional control, the most suitable manufacturing method is **Injection Molding**. Injection molding allows for the economical mass production of brush holders with consistent geometry, tight tolerances, and excellent surface finish. Additionally, it supports high-volume (for our case 100,000 units) for manufacturing with short cycle times, which is ideal for consumer-grade power tools. The choice of injection molding is further justified by its compatibility with engineering plastics that offer the necessary thermal resistance, structural rigidity, and electrical insulation required for safe and reliable motor operation.

Figure 11: Brush holder

The cost analysis for the brush holder was conducted using injection molding with Nylon 66, 30% glass fiber-filled material. The estimated production quantity was 100,000 units. The envelope dimensions were specified as 1 × 0.6 × 0.5 inches, with a maximum wall thickness of 0.1 inches and a projected area of 0.6 in², covering 100% of the envelope. The part volume was estimated at 0.3 in³, and high precision (≤ 0.005 inches) with smooth surface finish (Ra ≤ 32) was selected, reflecting the tight tolerances needed for proper fit and electrical insulation.

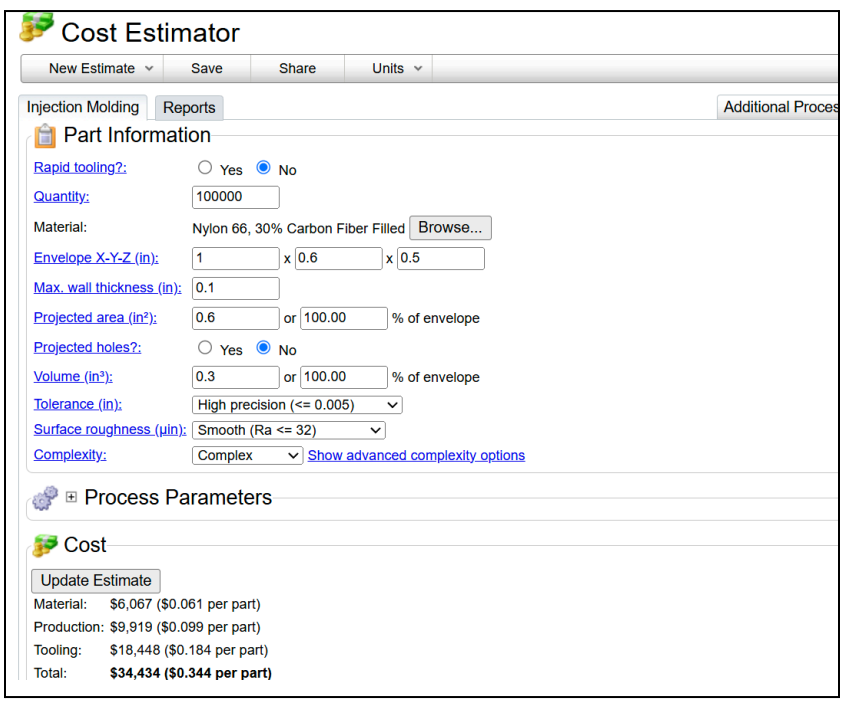

Figure 12: Cost Analysis of Injection Molded Brush Holder

This results in a total cost per part of $0.344 and a total project cost of $34,434 for 100,000 units. The relatively low material cost reflects the efficiency of injection molding for plastic components, while the tooling cost is significant upfront but amortized over the large production volume. The selection of a complex geometry setting also reflects the brush holder's intricate design, which houses the carbon brushes and ensures precise alignment with the motor assembly.

7. **Inner Flange:**

Its primary function is to secure non-hubbed grinding or cutting accessories directly to the grinder's spindle, ensuring proper alignment and safe operation. This inner flange is manufactured from **low carbon steel**, chosen for its strength, durability, and resistance to wear. Steel's mechanical properties make it ideal for withstanding the stresses and rotational forces encountered during grinding tasks. The production process for this component typically involves machining, such as turning and milling. These methods allow for tight tolerances and accurate dimensions, which are essential for ensuring a secure fit and reliable performance. **Precision machining** is favored for such parts because it provides the necessary accuracy and surface finish required for components that interact closely with other mechanical parts.

Figure 13: Inner flange

<u>Cost Estimation</u>:
The machining cost analysis for the inner flange was carried out using three primary CNC lathe operations: rough turning, finish turning, and drilling. The total setup time was calculated as 1.65 hours to account for tooling and machine preparation. For each part, the load/unload time totaled 57.2 seconds, and total tool positioning time across all operations was 12 seconds. The flange was machined from low carbon steel with a raw stock dimension of 2.0 inches in diameter and 0.5 inches in height, providing sufficient allowance for turning and facing.

In the first operation, rough turning was used to reduce the outer diameter from 1.88 inches to 1.8 inches over a length of 0.4 inches, removing 0.3642 in^3 of material. Using a surface generation rate of 29.95 in^2/min and a specific cutting energy of 1.1 hp·min/in^3, the machining time under recommended conditions was found to be 4.527 minutes, with 6.348 seconds added for extra tool travel. The second operation, finish turning, refined the diameter from 1.8 inches to 1.75 inches over the same length, removing 0.1393 in^3. This required 8.581 minutes due to the lower surface generation rate (15.36 in^2/min), with an additional 6.562 seconds of tool travel. The final operation, drilling, was performed to create the center hole (from 1.75 in to 0.628 in) and removed 2.0945 in^3 of material.

Table 1: Machining Cost Analysis for Inner Flange

	Operations	Setup Time (hr)	Load/Unload Time (sec)	Tool Positioning (sec)	Dimension lw (in)	Dimension Da (in)	Dimension Db (in)	Volume (in3)	Specific Cutting Energy (hp mm/in3)	Available Power (hp)	Machinging Time Max Power	Rate of Surface Generation (in2/min)	Area (in2)	Machining Time Recommended Condition	Time for extra tool travel t'_m
1	Rough Turn	0.65	23.3	1.5	0.4	1.88	1.8	0.3642	1.1	8	2.9898	29.95	2.2608	4.527	6.348
2	Finish Turm			1.5	0.4	1.8	1.75	0.1393	1.1	8	1.1492	15.36	2.198	8.581	6.562
3	Drilling	1	33.9	9	0.4	1.75	0.628	2.0945	0.95	2	59.69	27.39	2.198	4.814	2.491
	Total	1.65	57.2	12											15.401

The total cost per part for manufacturing the inner flange was estimated by considering all key contributors: material, setup, non-productive, and machining costs. The material cost for the part, based on a low carbon steel workpiece weighing approximately 0.4461 lb, was calculated to be $0.60 per part. The setup cost, which accounts for the 1.65 hours required to prepare the CNC lathe and tooling, was distributed across the production volume and calculated as $0.396 per part. Non-productive cost, including load/unload time and tool positioning across all three operations, contributed $0.577 per part. Finally, the machining cost, which includes rough turning, finish turning, and drilling based on specific cutting energy and material removal volume, was estimated to be $0.1598 per part. From the calculated cost, we get total cost to be $1.7398,

8. Shoulder Pin

This pin plays a crucial role in facilitating the engagement and disengagement of the spindle lock mechanism, allowing users to safely and efficiently change grinding or cutting accessories. Crafted from hardened **Alloy Steel AISI 4140**, the shoulder pin is designed to endure the mechanical stresses associated with frequent use. The manufacturing process involves **precision machining** techniques, such as turning and milling, to achieve the exact dimensions and tolerances required for proper function. Subsequent heat treatment is applied to attain the desired hardness and durability. These manufacturing steps are essential to ensure the shoulder pin's reliability and performance in demanding applications.

Figure 14: Shoulder pin

Cost Estimation

The machining cost estimation for the shoulder pin includes three turning operations: two rough cuts and one finish cut, all performed on a CNC turret lathe using alloy steel. A setup time of 0.65 hours was used to account for machine preparation and tool setup. The load and unload time per part was taken as 10.3 seconds based on Table 7.3 (Textbook), which corresponds to small parts held in a collet. Each operation also includes 1.5 seconds of tool positioning time, which reflects turret indexing between tools.

The first rough turning assuming material removal from the diameter from 0.7in (initial workpiece) to 0.6 in over a 1-inch length. The second rough turning step reduces the diameter further from 0.6 in to 0.25 in over 0.4 inches this step was added to reflect an additional feature on the part after the chamfer. The final operation is a finish turn from 0.6 in to 0.5 in over 1 inch to improve surface finish and dimensional accuracy.

Table 2: Machining cost analysis for shoulder pin.

	Material	Alloy Steel													
	Machine	CNC Turret lathe													
	Operations	Setup Time (hr)	Load/Unload Time (sec)	Tool Positioning (sec)	Dimension lw (in)	Dimension Da (in)	Dimension Db (in)	Volume (in3)	Specific Cutting Energy (hp mm/in3)	Available Power (hp)	Machinging Time Max Power	Rate of Surface Generation (in2/min)	Area (in2)	Machining Time Recommended Condition	Time for extra tool travel t_m'
1	Rough Turn	0.65	10.3	1.5	1	0.7	0.6	0.4661	1.3	8	4.127	112.32	1.884	1.005	6.03
2	Rough Turn			1.5	0.4	0.6	0.25	0.3736	1.3	8	3.6426	112.32	0.314	0.167	5.21
3	Finish Turn			1.5	1	0.6	0.5	0.1885	1.3	8	1.837	57.6	1.57	1.635	6.47
	Total	0.65	10.3	3											17.71

Total machining cost that was obtained from calculation which included material cost ($1.005 per par) assuming price of alloy steel is $1.5/lb, setup cost ($0.016) per part, non-productive cost to be ($0.11 per part) and machining cost with allowances was $0.147 per part. So, the total cost was estimated to be $1.278 per part assuming manufacturing of 100,000 units.

9. Spindle:

The spindle in the angle grinder is a central rotating component that connects the motor's output to the grinding or cutting wheel. It ensures the secure attachment and proper rotation of various accessories, such as grinding wheels, cut-off discs, and wire brushes. This spindle is typically manufactured from **alloy steel**, chosen for its high strength, durability, and resistance to wear. The hardening process enhances the steel's ability to withstand the stresses and rotational forces encountered during grinding operations, ensuring a long service life. The production of the spindle involves **precision machining** processes, such as turning and milling, to achieve the exact dimensions and tolerances required for proper fit and function. After machining, the spindle undergoes heat treatment to attain the desired hardness and strength characteristics. These manufacturing steps are crucial to ensure the spindle's reliability and performance in demanding applications.

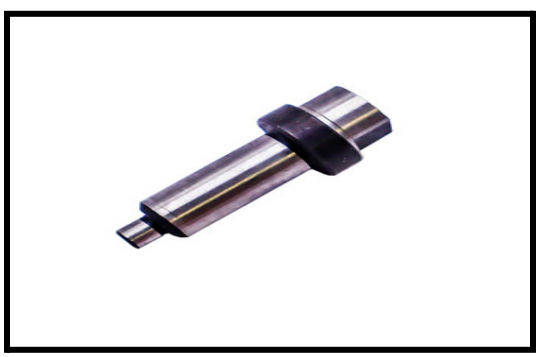

Figure 15: Spindle

Cost Estimation:

The spindle was manufactured using four turning operations on a CNC lathe: three rough turning steps followed by a finish turning pass. The total setup time for all operations was recorded as 0.65 hours, accounting for initial tool setup and machine preparation. The load/unload time per part was estimated at 10.3 seconds, and tool positioning time totaled 6 seconds across all passes. In the first rough turning operation, the part was reduced from 1.0 inch to 0.88 inch over a length of 1.95 inches, removing 0.3453 in^3 of material. This was followed by two additional rough cuts: one reducing the diameter from 0.88 in to 0.25 in over 0.3 inches removing 0.6706 in^3 and another from 0.88 in to 0.5 in over 0.4 inches (removing 0.6586 in^3). Each rough pass used a specific cutting energy of 1.3 hp·min/in^3 and available power of 8 hp to calculate machining time.

The finish turning operation brought the part to its final geometry over the full 1.95-inch length, removing 0.8229 in^3. Machining time under recommended cutting conditions was also calculated based on the surface generation rate of 112.32 in^2/min and the contact area for each operation. These values yielded respective machining times of 2.878 min, 0.126 min, 0.335 min, and 2.779 min for the four operations. Finally, tool approach and retract time was added to each operation, totaling 24.05 seconds. This analysis provides an accurate estimate of machining time and workload distribution for the spindle, enabling cost estimation for high-volume production.

Table 3: Machining Cost Analysis for Spindle

	Operations	Setup Time (hr)	Load/Unload Time (sec)	Tool Positioning (sec)	Dimension lw (in)	Dimension Da (in)	Dimension Db (in)	Volume (in3)	Specific Cutting Energy (hp mm/in3)	Available Power (hp)	Machinging Time Max Power	Rate of Surface Generation (in2/min)	Area (in2)	Machining Time Recommended Condition	Time for extra tool travel t'_m
1	Rough Turn	0.65	10.3	1.5	1.95	1	0.88	0.3453	1.3	8	3.366	112.32	5.388	2.878	6.44
2	Rough Turn			1.5	0.3	0.88	0.25	0.6706	1.3	8	6.538	112.32	0.236	0.126	5.56
3	Rough Turn			1.5	0.4	0.88	0.5	0.6586	1.3	8	6.42135	112.32	0.628	0.335	5.61
2	Finish Turn			1.5	1.95	0.88	85	0.8229	1.3	8	8.023275	112.32	5.204	2.779	6.44
	Total	0.65	10.3	6											24.05

Based on the calculated machining times, the total cost of manufacturing the spindle was determined by breaking down setup, non-productive, machining, and material costs. The setup cost, assuming an operator rate of $30 per hour, was calculated to be $0.016 per part. The non-productive cost, which includes loading/unloading and tool positioning, amounted to $0.1358 per part. The actual machining cost, derived from cutting times and machine rates, was estimated at $0.20 per part. For material, using 4140 alloy steel priced at $1.50 per pound, the material cost per part came out to $1.00. Altogether, these elements result in a total cost of $1.352 per part.

10. Switch Lever

The switch lever used in the Makita 9557NB angle grinder is a crucial component that enables the user to control the tool's power state. It interfaces with the switch to turn the grinder on or off, and may also feature a lock-on function for continuous operation. This component is typically manufactured from high-strength thermoplastic, such as **polyamide (nylon 66, 10% glass filled)**, chosen as it provides improved stiffness and dimensional stability over unreinforced nylon, without making the part overly rigid or brittle., impact resistance, and ability to withstand the mechanical stresses of repeated use. Thermoplastics are also resistant to various environmental factors, including moisture and temperature fluctuations, making them suitable for power tool applications. The production process for the switch lever involves **injection molding**, a method where molten plastic is injected into a mold to form the desired shape. Injection molding is ideal for producing complex geometries with high precision and

repeatability, which is essential for components like switch levers that require accurate dimensions to function correctly within the tool's housing. Additionally, this process is cost-effective for mass production, ensuring consistency and quality across large quantities.

Figure 16: Switch Lever

Cost Estimation:

The cost estimation for the switch lever component of the angle grinder was performed using an injection molding process with a production volume of 100,000 units. The material selected was Nylon 66, 10% Glass Fiber Filled, a common engineering thermoplastic used in mechanical actuators due to its excellent balance of strength, dimensional stability, and wear resistance. Although the cost estimator interface displays ABS, the geometric and cost input methodology remains applicable and reflective of the selected nylon grade.

The envelope size was specified as 9 × 6 × 0.4 inches in product description (amazon), which represent the maximum extents of the switch lever, including its mounting base and actuation profile. These values were derived from actual product references and represent the envelope that the part would occupy in a mold cavity. The maximum wall thickness was set to 0.2 inches, matching standard practice for injection molded levers that require rigidity but also fast and even cooling. The projected area of 54 in^2 corresponds to the top-view surface area of the component, important for determining injection pressure and mold design. Volume was calculated at 10.73 in^3 based on the known part weight (0.5 lb) and the density of glass-filled nylon.

Tolerances were set to high precision (≤ 0.005 in) to ensure consistent fitment with the switch mechanism and housing, and surface roughness was designated as Smooth (Ra ≤ 32) to provide a comfortable tactile finish for the user's finger while maintaining mechanical quality. The complexity was selected as Simple, as the switch lever is a single-shot part without undercuts or sliding cores, featuring fewer than 10 total features.

The final cost breakdown showed a material cost of $77,485 ($0.775 per part), a production cost of $88,581 ($0.886 per part), and a tooling cost of $44,626 ($0.446 per part). The total cost for 100,000 parts was $210,691, or $2.107 per part. These values reflect the economy of scale achieved in mass production, where tooling costs are amortized and material and cycle efficiencies are maximized. This analysis confirms that injection molding is a cost-effective and functionally appropriate method for producing the switch lever in large quantities.

11. Switch knob:

The angle grinder utilizes the switch knob to control its power functions. This component is designed for durability and ease of use, ensuring reliable operation during various grinding tasks. It facilitates the on/off operation of the angle grinder, providing a tactile interface for the user. It is typically constructed from high-strength thermoplastic, such as **30% glass fiber-reinforced nylon (PA6-GF30)**, offering durability and heat resistance. The manufacturing process used is **Injection Molding** due to its repeatability and cost-effectiveness for mass production.

Figure 17 - Switch knob

Cost estimation:

The dimension of the switch know is (1*1*1) inches or (25.4*25.4*25.4)mm. Considering it is a very small part to manufacture, the projected area is 1 in^2 and also the projected volume is 1 in^3. Assuming high precision, smooth surface roughness and moderate complexity of the switch knob and defect rate of 5%. From custompart.net, we got a standard machine size of 30 tons which has an hourly rate of $25 and assuming 95% uptime of the machine for single cavity mold, the cost for one part came out to be $0.395 as shown in the figure. The material cost is $15,045 ($0.150 per part), production cost is $7460 ($0.075 per part), tooling cost is $17,034 ($0.170 per part).

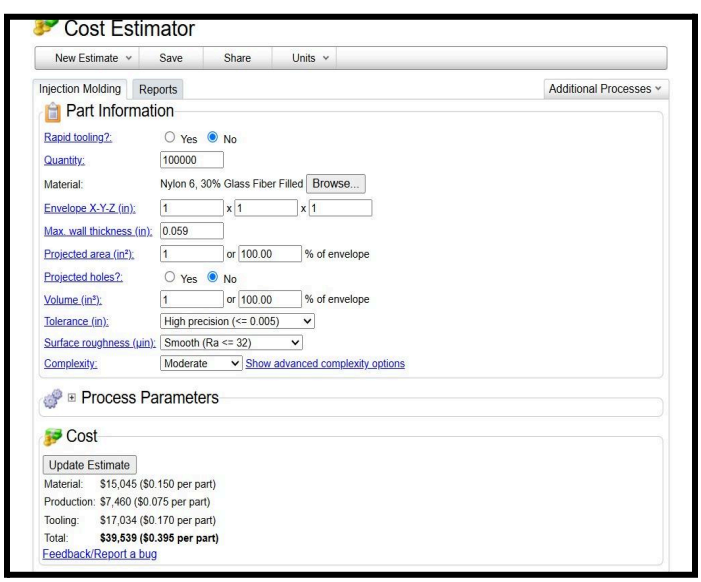

Figure 18 - Switch knob cost

12. Rear cover:

The rear cover of the angle grinder is the protective casing located at the back of the grinder, enclosing the brush holder, power cord entry, and electrical terminals. It's typically made from durable plastic and forms the rear grip section of the tool. It protects internal components such as the brush holder, wiring, and rear end of the armature, **s**upports ventilation, often featuring slots or grills to allow airflow drawn by the cooling fan, houses the power cord entry, provides access to motor brushes for inspection or replacement. The rear cover may seem like a simple part, but it's critical for the safety, cooling, maintenance, and usability of the angle grinder. It enables smooth operation and long-term reliability while contributing to user comfort and tool serviceability.

Figure 19 - Rear cover

The material used is the same as the switch knob which is **PA6-GF30 (Nylon 6 with 30% Glass Fiber Reinforcement)**. It has excellent mechanical strength and impact resistance, high heat resistance, electrical insulation and good dimensional stability. Alternative materials like ABS and polyethylene terephthalate (30% glass) could be used but they have low heat resistance and cost of the material is slightly higher than nylon 6. The manufacturing process used is **Injection Molding** as it is cost-effective for manufacturing 100,000 products per year.

<u>Cost estimation</u>:

The diameter of the part is 59 mm, length is 60 mm and assuming wall thickness to be 4 mm. The projected area came out to be 4.23 in^2, and the projected volume is 9.98 in^3. The rear cover of the grinder has 18 slots and 2 small holes for ventilation purposes. The total projected area of the holes is assumed to be 0.785 in^2 (assuming 1 inch diameter including the holes and the slots). The tolerance level is <= 0.005 in or 0.127 mm (high precision), smooth surface roughness and moderate complexity.

The size of the machine is 80 tons with an hourly rate of $80 according to the custompart.net. Assuming 95% machine uptime and single cavity mold, the material cost came out to be $148,963 ($1.49 per part), production cost is $25,760 ($0.258 per part), tooling cost is $26,438 ($0.264 per part). The total cost per part came out to be $2.012 as shown in the figure.

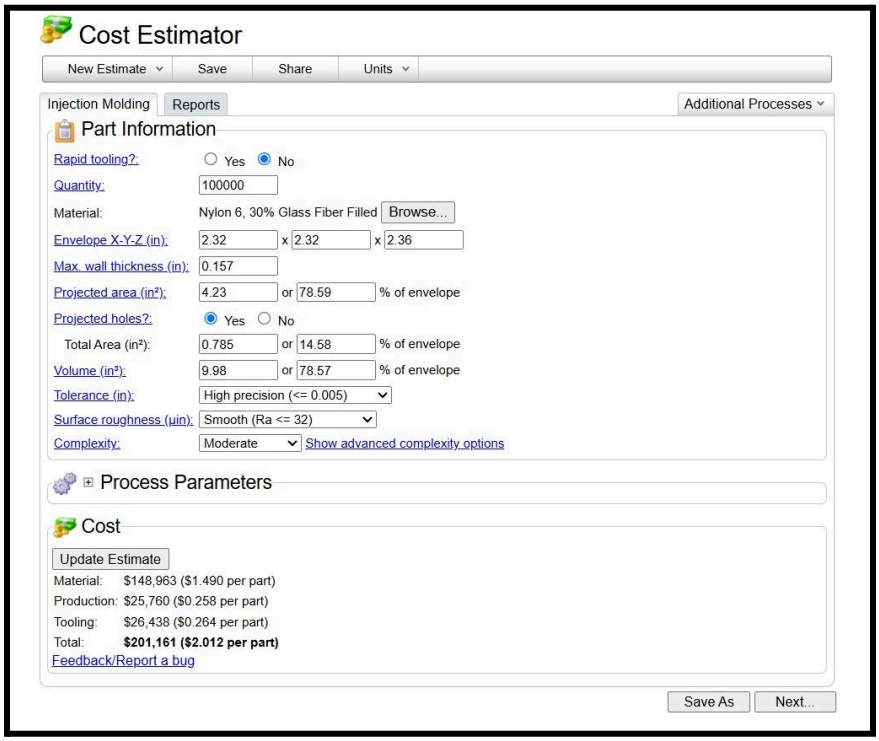

Figure 20 - Rear cover cost

13. Baffle plate:

The baffle plate in an angle grinder is a crucial internal component that directs cooling air over the motor, helping to regulate temperature and protect internal parts from overheating. It also prevents dust and debris from entering sensitive areas like the motor and bearings. This enhances the tool's durability, performance, and overall lifespan.

Figure 21 - Baffle plate

The material used is **nylon 6, 30% glass-fiber filled**. The cost per kg of the material is $2.15 as per custompart.net. Nylon 6 is heat resistant, light weight, corrosion resistant, good electrical insulation, and has good impact resistance. The manufacturing process used to manufacture this part is **Injection Molding** as it is cost-effective for mass production.

Cost calculation:

The outer diameter of the baffle plate is assumed to be 50 mm, height of 20 mm and maximum wall thickness is 2 mm. The projected area came out to be 3.02 in^2, projected volume is 0.23 in^3. There is a center bore and 2 mounting holes on the edge, hence the projected area is 0.05 in^2. Assuming high precision tolerance, smooth surface roughness and moderate complexity. The size of the machine is 17 tons and an hourly rate of $25, 95% machine uptime, and single cavity mold. The material cost per part is $0.025 per part, production cost is $0.163, tooling cost is $0.222. The total cost per part came out to be $0.410 as shown in the figure.

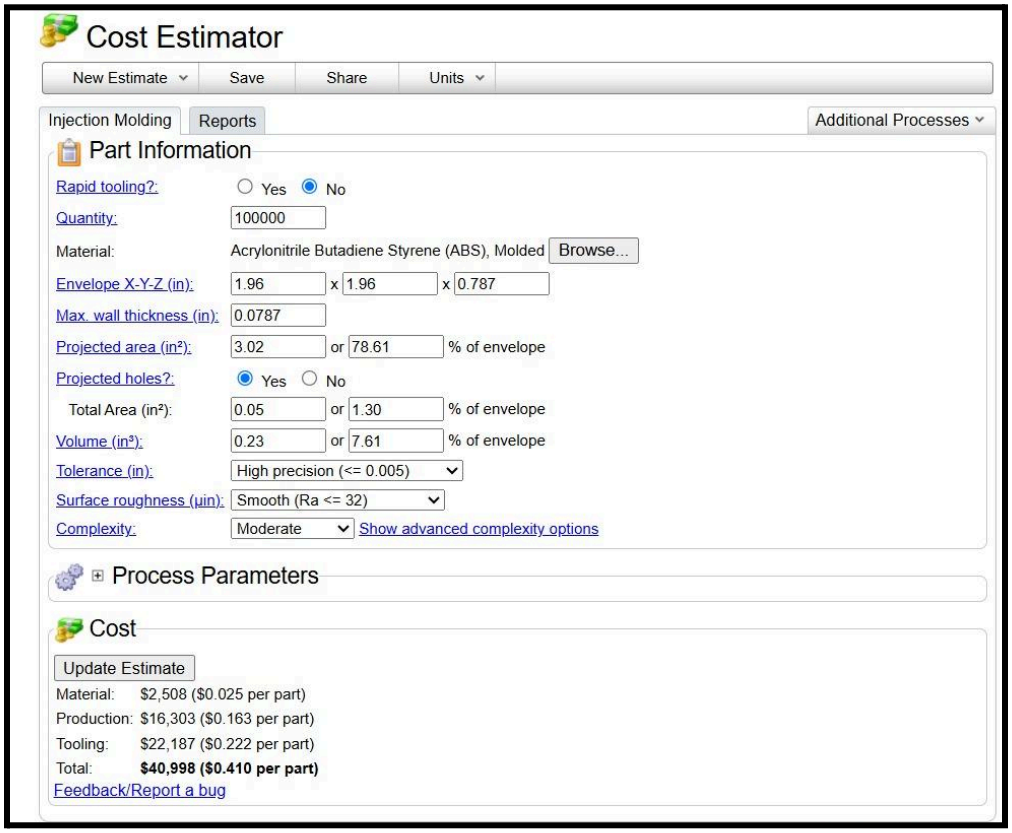

Figure 22 - Baffle plate cost.

Cost Analysis of Standard Parts:

Part	Quantity	Material	Cost per part (in $)
Ball Bearing	1	Steel	0.89
Insulation Washer	1	Steel	0.34
Armature Assembly	1	Plastic	4.23
Hex nut M6	1	Steel	0.62
O-Ring 26	1	Steel	0.20
Ball Bearing 629LLB	1	Steel	0.89
Pin Cap	1	Rubber	0.66
Compression Spring 8	1	Steel	0.12
Tapping Screw 4x30	4	Steel	0.21
O Ring	1	Steel	0.12
Rubber Pin	1	Plastic	0.73
Ball Bearing 696ZZ	1	Steel	0.89
Retaining Ring S-12	1	Steel	0.74
Wave Washer	1	Steel	0.82
Retaining Ring R-32	1	Steel	0.12
Ball Bearing 6201 DDW	1	Steel	0.89
Flat Washer 12	1	Steel	0.34
O Ring	1	Steel	0.12
P.H Screw M4X14	4	Steel	0.50
Labyrinth Ring	1	Steel	0.62
WOODRUFF KEY 4	1	Steel	0.21
Lock Nut ⅝-45	1	Steel	0.74
Tapping Screw 4 X 80	2	Steel	0.80
Tapping Screw 3x10	4	Steel	0.34

Carbon Brush Set	1	Carbon	1.34
Spacer	1	Plastic	0.12
Strain Relief	1	Plastic	1.2
Tapping Screw 4X18	2	Steel	0.80
Cord Guard	1	Copper + PVC	1.2
Cord	1	Copper + PVC	2.3
TAPPING SCREW PT 4X18	1	Steel	0.72
Wheel Cover	1	Steel	0.34
Total			**24.16**

Design for Manufacturing (DFM):

The hand grinder assembly applies several key Design for Manufacturing (DFM) principles to improve manufacturability and reduce cost. Below are three key DFM principles applied to the parts in this assembly, along with specific examples:

1. **Part Complexity:** For example, the plastic rear handle casing may currently have undercuts to accommodate features like snap-fits, cable channels, or decorative contours. While these undercuts may help functionally or aesthetically, they significantly increase manufacturing complexity because they require complex injection mold tooling with side actions, lifters, or collapsible cores. By redesigning for example, replacing snap-fits with simpler screw joints or redesigning cable channels to follow the draft direction—the part could be manufactured using a straight-pull two-part mold.
2. **Uniform thickness-** The hand grinder assembly demonstrates adherence to the DFM principle of uniform wall thickness in several injection-molded and stamped components, ensuring manufacturability and structural consistency. For example, the gear housing and motor housing maintain uniform thickness in their plastic sections to prevent defects like warping or sink marks during molding. This uniformity aids in mold flow efficiency, reduces cycle times, and minimizes material waste, aligning with DFM goals for cost-effective and high-quality manufacturing.
3. **Material Choice Based on Function:** The assembly follows the DFM principle of selecting materials that match the functional requirements of each part, ensuring optimal performance, durability, and cost-effectiveness. For example, the spindle is made of 4140 alloy steel, chosen for its high strength and fatigue resistance, which are critical for transmitting rotational motion under load. The inner flange uses low carbon steel, providing sufficient strength and machinability at a lower cost, as it mainly functions to secure the grinding wheel. The brush holder is made from PA66 nylon with glass fiber reinforcement, which offers excellent electrical insulation and thermal stability, critical for safely housing the carbon brushes near the motor. By carefully matching materials to each part's mechanical, thermal, and electrical requirements, the assembly achieves reliable performance while minimizing material waste and manufacturing cost.

Design for Assembly (DFA):

Design for Assembly (DFA) is an engineering approach focused on simplifying product structure to make assembly easier, faster, and more cost-effective. It involves minimizing the number of parts, using standard components, and designing features that ensure easy handling, alignment, and fastening. By optimizing a product for efficient assembly, DFA reduces labor costs, assembly time, and the potential for errors, ultimately improving overall manufacturing efficiency and product reliability.

1. Consolidating parts:
The baffle plate and the field of the angle grinder can be consolidated, resulting into one single part. The current assembly from the exploded view requires 4 parts which are a field, baffle plate and two screws to mount the plate on the field. If we redesigned the parts then one single part containing both baffle plate and the field can be manufactured and hence save manufacturing costs and assembly time. The standard screws are eliminated which results in faster assembly of the overall product. The redesigning of the field requires creating an extrusion cut on the field (basically a slot) so that the baffle plate can have a snug fit and lock both the parts together.
2. Pre-assembled subassemblies:
Pre-assembling of subassemblies include assembly of standard and non-standard parts prior to the main assembly of the entire product. The main advantage of this is that the entire subassembly can be treated as one part and could reduce the assembly time drastically. Some of the examples where it could be very beneficial in the hand grinder is:

1. The armature assembly along with fan, labyrinth rubber ring, ball bearings, insulation washer, o-ring, and spiral pinion can be pre-assembled in the gear housing cover. This would significantly reduce the assembly time and they can be treated as one assembly part. This would reduce the assembly part count from 10 to 1.
2. Similarly, this can be done to the spiral gear assembly. We can integrate spiral gear, bearings, o-ring, washer, retaining ring into the bearing box. This prior assembly of standard and non-standard parts will be very beneficial in reducing down the handling and warehousing costs

3. Reducing number of parts: Reducing the number of fasteners lowers assembly time, reduces part costs, and improves maintainability. Parts like the baffle plate and pin cap appear to use snap-fit or press-fit mechanisms instead of screws or nuts. Even when fasteners are necessary, such as the tapping screws used for the gear housing cover, they are used sparingly and strategically, demonstrating a clear effort to minimize the total fastener count and streamline assembly.

Item no	Item name	Quantity	Manual handling code	Handling time (s)	Manual Insertion code	Insertion time (s)	Total operation time (s)	Minimum no of parts
1	Field	1	30	1.95	00	1.5	3.45	1
2	Baffle plate	1	33	2.51	38	6	8.51	0
3	Labyrinth ring	2	41	6.85	00	1.5	8.35	0
4	Ball bearing	4	00	1.13	00	1.5	2.63	4
5	Insulation ring	1	41	6.85	00	1.5	8.35	0
6	Armature assembly	1	00	1.13	06	5.5	6.63	1
7	Fan	1	11	1.8	00	1.5	3.3	1
8	Gear housing cover	1	10	1.5	01	2.5	4	1
9	O-ring	2	41	6.85	00	1.5	8.35	0
10	Spiral gear	2	00	1.13	08	6.5	7.63	2
11	Gear housing	1	10	1.5	39	8	9.5	1
12	Compressi on spring	2	98	7	18	9	16	0
13	Tapping screw	9	01	1.43	38	6	7.43	0
14	Shoulder pin	1	10	1.5	08	6.5	8	0
15	Rubber pin	1	41	6.85	08	6.5	13.35	0
16	Wave washer	1	41	6.85	00	1.5	8.35	0
17	PH screw	5	01	1.43	38	6	7.43	0
18	Bearing box	1	10	1.5	06	5.5	7	1

19	Woodruff key	1	60	4.8	08	6.5	11.3	0
20	Spindle	1	10	1.5	00	1.5	3	1
21	Wheel cover	1	30	1.95	38	6	7.95	0
22	Inner flange	1	00	1.13	06	5.5	6.63	1
23	Lock nut	2	00	1.13	35	7	8.13	0
24	Name plate	1	00	1.13	38	6	7.13	0
25	Brush holder	1	21	2.1	00	1.5	3.6	1
26	Carbon brush set	1	01	1.43	06	5.5	6.93	1
27	Switch knob	1	02	1.8	00	1.5	3.3	1
28	Motor housing	1	30	1.95	08	6.5	8.45	1
29	Rear cover	1	10	1.5	00	1.5	3	0
	Total	**49**					**207.68**	**18**

Estimated assembly time= **207.68 s**
Design efficiency= (18*3)/207.68=0.2600= **26%**
Assuming assembly cost for all part to be 3 cents to maintain uniformity,
Assembly cost= 207.68*3 = 623.04 cents= **$6.23**

Cost analysis for overall design:

The overall cost analysis for the hand grinder assembly shows a total estimated cost of $83.91. This cost is broken down into three main categories. The non-standard parts, which include custom components built in-house such as the spindle, inner flange, brush holder, and gear housing, account for the largest portion at $53.52. These parts typically require specialized materials, processes, or tooling, which drives up their contribution to the total cost. The standard parts, including fasteners, bearings, and screws, contribute $24.16, reflecting the benefit of using off-the-shelf components to control costs due to discount on bulk purchase from vendors.. Finally, the assembly cost is calculated at $6.23, which includes labor, equipment, and overhead associated with putting together the complete product. This breakdown highlights that while non-standard parts dominate the cost structure, opportunities exist to optimize both material and assembly strategies to improve cost efficiency in future designs.

Table 4: Cost Analysis for Overall Design of Assembly

Parts	Cost ($)
Non-standard parts	53.52
Standard parts	24.16
Assembly	6.23
Total	**83.91**

Conclusion:

The comprehensive evaluation and redesign of the hand grinder using the principles of Design for Manufacturing (DFM) and Design for Assembly (DFA) highlight the transformative potential of these methodologies in improving both product design and manufacturing efficiency. Through a detailed analysis of each component-including the gears, spindle, casings, fasteners- this project demonstrates how targeted design changes can significantly reduce manufacturing costs, simplify assembly, and enhance overall product performance.

The DFM approach enabled the identification of parts and features that were either overly complex or inefficient to produce. By revisiting material selections and simplifying component geometries, the manufacturing process was streamlined to reduce machining time, tooling requirements, and material waste. For example, replacing multi-step machining operations with features conducive to molding or simpler turning operations contributed to cost reductions without compromising the grinder's functionality. Additionally, optimizing tolerances helped maintain the necessary precision while minimizing the use of expensive manufacturing techniques.

Simultaneously, the application of DFA principles revealed opportunities to reduce the total number of parts and simplify their interconnections. Redesigning components to serve multiple functions or integrating subassemblies led to a more efficient and intuitive assembly process. The reduction of fasteners and the incorporation of snap-fit or interference-fit features lowered the reliance on manual labor, reducing assembly time and associated labor costs. Improved alignment features further minimized errors during assembly, thereby increasing consistency and product quality.

The cost estimation analysis comparing the original and optimized designs confirmed the tangible benefits of the redesign. Both direct manufacturing costs and labor-related assembly expenses saw measurable reductions. In many cases, these savings were accompanied by improvements in ease of serviceability and durability, thus enhancing the overall value proposition of the product. The optimization also contributed to more sustainable manufacturing practices through reduced material use and waste generation.

Beyond the specific improvements achieved for the hand grinder, this project establishes a replicable design and evaluation methodology that can be extended to a wide range of mechanical consumer products. In increasingly competitive markets, manufacturers must continually seek ways to offer high-performance products at lower prices without compromising quality or functionality. The disciplined application of DFM and DFA principles, as demonstrated in this study, offers a strategic path toward achieving those objectives.

In conclusion, the integration of DFM and DFA principles into the product development process has proven to be an effective strategy for enhancing manufacturing efficiency, reducing production costs, and improving assembly processes. The insights gained from this project reinforce the value of early-stage design optimization and provide a foundation for continuous improvement in product design.

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